MIL-M-62173(AT)
24 May 1973
SUPERSEDING
DAPD-268A
12 May 1961

MILITARY SPECIFICATION

MUFFLER, EXHAUST

1. SCOPE

1.1 Scope. This specification establishes the requirement for one type of exhaust muffler (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-T-830	-	Tube, Steel, Carbon, Mechanical, Round: Seamless and Welded.

Military

MIL-C-490	-	Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings.
MIL-S-4174	-	Steel, Sheet and Strip, Flat, Aluminum Coated, Low Carbon.
MIL-E-5272	-	Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-P-14276	-	Paint, Heat Resisting; Silicon, Aluminum.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-1261	-	Welding Procedures for Constructional Steels.

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DRAWINGS
Ordnance
D9098035 - Muffler, Exhaust.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. Prior to acceptance of production mufflers, a preproduction sample of mufflers shall be furnished for testing as specified in 4.2.

3.2 Materials. The main body of the muffler shall be made from Type I, grade A, 16 gage aluminized carbon steel conforming to Specification MIL-S-4174. The outlet tube shall be made from either aluminized steel as specified above or from carbon steel tube, Composition MT 1010 to MT 1020, condition optional, 16 gage conforming to Specification QQ-T-830.

3.3 Construction. Construction of the muffler shall be in accordance with the requirements of this specification and Drawing D90908035. Components of the muffler not specifically described herein, shall conform to high standards of quality and be of the lightest feasible weight. Welding shall be in accordance with Specification MIL-STD-1261 Class 1.

3.3.1 Finish. When carbon steel tube conforming to Specification QQ-T-830 is used, the exterior surfaces shall be finished as follows:

(a) Sand blast in accordance with Specification MIL-C-490, grade II, Type 1.
(b) Immediately after sand blasting, apply one coat of aluminum pigmented silicone paint conforming to Specification MIL-P-14276. Dry film thickness shall be 0.8 to 1.0 mil.

3.4 Performance and product characteristics.

3.4.1 Pressure drop. The maximum pressure drop through the muffler shall now exceed 20 inches of water at engine rated power and speed (see 4.4.4).

3.4.2 Noise level. The muffler shall reduce the over-all noise of exhaust gases to a maximum of 100 decibels (db.) at any point on a radius of 10 feet from the end and 2 feet above the muffler tail pipe, with engine operating at rated load and speed (see 4.4.5).

3.4.3 Shock. After being tested as specified in 4.4.2 the muffler shall meet the performance requirements of 3.4.1 and 3.4.2.

3.4.4 Vibration. After being tested as specified in 4.4.3 the muffler shall meet the performance requirements of 3.4.1 and 3.4.2.

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3.5 Product marking. Markings shall be in accordance with Standard MIL-STD-130 and shall be located as specified on Drawing D9098035. The date of manufacture shall be legibly stamped on the muffler.

3.6 Workmanship. Workmanship shall be such as to produce mufflers free of nicks, burrs, scratches, tool marks or other defects which may adversely affect performance, product strength, endurance, or appearance of the unit. Welding shall be complete and uniform.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction testing. Unless otherwise specified, a preproduction sample of 2 specimens shall be submitted for preproduction testing. The sample shall be representative of the mufflers proposed to be furnished to the Government. Preproduction testing shall be conducted by the Government at a place designated by the Government and shall consist of examination for the defects specified in 4.3.3 and testing as specified in table I in the order listed.

Table I. Order of preproduction testing

Specimen 1	Specimen 2
Shock (4.4.2)	Vibration (4.4.3)
Pressure Drop (4.4.4)	Pressure Drop (4.4.4)
Noise Level (4.4.5)	Noise Level (4.4.5)

4.2.1 Failure. Failure of a preproduction sample to pass any of the examinations or tests specified herein may be cause for the Government to refuse to conduct additional examination or testing until the faults revealed have been corrected.

4.3 Acceptance inspection.

4.3.1 Lot. Unless otherwise specified, a lot shall consist of all mufflers from an identifiable production period, submitted at one time for acceptance.

4.3.2 Sampling.

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4.3.2.1 For acceptance examination. Samples for acceptance examination shall be selected in accordance with inspection level II of Standard MIL-STD-105.

4.3.2.2 For acceptance testing. Samples for acceptance testing shall be selected in accordance with inspection level L 8 of Standard MIL-STD-105.

4.3.3 Acceptance examination.

4.3.3.1 Acceptable quality level. Each sample selected in accordance with 4.3.2.1 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.3.2 Classification of defects. Examination shall be conducted as specified in table II. Any mufflers in the sample containing one or more defects shall be rejected.

Table II. Classification of defects in accordance with standard MIL-STD-105

Categories	Defects
Critical:	None Defined
Major	
101	Not in conformance as to material (see 3.2)
102	Not in conformance as to construction (see 3.3)
103	Physical damage
Minor	
201	Marking incomplete or missing (see 3.6)
202	Poor workmanship (see 3.7)

4.3.4 Acceptance testing. Samples selected in accordance with 4.3.2.2 that have passed the examination specified in 4.3.3 shall be subjected to the tests specified in 4.4.4 and 4.4.5 using an AQL of 6.5 on the basis of percent defective.

4.4 Test procedures.

4.4.1 Test temperature requirements. Unless otherwise specified, the unit shall be tested at room temperature, 77 degrees plus or minue 18 degrees F.

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4.4.2 Shock. To determine conformance to 3.4.3 the unit shall meet the requirements of the shock test in accordance with Specification MIL-E-5272, Procedure V, except that the "Equipment Crash - Safety" test shall be omitted.

4.4.3 Vibration. To determine conformance to 3.4.4 the unit shall be subjected to the following vibration test procedures:

Test	Frequency Range	Acceleration	Time Duration Room Temp.
Resonance	10-52 cps 52-500	$\pm$ 4g $\pm$ 5g	60 Min.
Cycling	10-500-10 cps	Same as above	Four 15-Min. cycles

(a) The tests shall be performed consecutively along 3 mutually perpendicular axes of the unit.
(b) The unit shall be inspected for damage and shall meet the functional tests of paragraph 4.4.4 and 4.4.5 specified herein immediately after each test.
(c) The times spent at any resonant mode shall be not less than half as long as those given in table above; if no resonance is encountered, the resonance test times shall be twice the above times for a sweep through the entire frequency range.
(d) An exponential sweep rate shall be used whenever possible.
(e) During the resonance and cyling tests the test specimen will not be in operation.

4.4.4 Pressure drop. To determine conformance to 3.4.1, hot gases from an internal combustion engine (see 6.1) or an apparatus arranged to simulate an engine shall be put into the muffler and the total pressure (static plus dynamic) at the intake and exhaust ends of the muffler shall be measured. The difference in pressure through the muffler shall not exceed 20 inches of water at engine rated load and speed.

4.4.5 Noise level. To determine conformance to 3.4.2 the muffler shall be attached to the engine (see 6.1), and the engine operated at rated load and speed. The noise level shall be measured at a radius of 10 feet from the end and 2 feet above the end of the muffler tail pipe. The overall noise of the engine exhaust gases shall not exceed 100 db.

4.5 Inspection of the preservation, packaging, and packing. The preservation, packaging, and packing shall be inspected for conformance to the instructions in section 5, and to the specifications referenced therein.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging shall be level A or C, and packing shall be level A, B, or C as specified (see 6.2). Cleaning, preservation, packaging, packing and marking shall be in accordance with the applicable packaging data sheet or packaging standard as furnished by the responsible Government activity, at the time of invitation for bids.

6. NOTES

6.1 Intended use. This muffler is intended to be used on a 4-cycle, 4-cylinder (in-line) internal combustion gasoline engine of approximately 70 horsepower.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Quantity of preproduction samples required (see 4.2).
(c) Special markings required (see 5.3).
(d) Lot size for acceptance examination (see 4.3.1).
(e) Whether packaging data sheet will be furnished (see 5.1).
(f) Level of preservation and packaging and level of packing (see 5.1).

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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